



COURSE OUTLINE

COURSE CODE: 3666-018

COURSE TITLE: Applied Natural Language Processing

INSTRUCTOR: Saeid Abolfazli (PhD) | Mahmudul Hasan (MMSc)

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DAY/TIME:

September 16, 2024 to December 9, 2024

WEBINAR SCHEDULE: 6:00 pm – 9:00 pm

REQUIRED TEXT(S):

Ref 1: Benjamin Bengfort, Rebecca Bilbro & Tony Ojeda (2018) Applied Text Analysis with Python: Enabling Language-Aware Data Products with Machine Learning, O'Reilly, 978-1491963043

RECOMMENDED TEXT(S) AND OTHER RESOURCES:

Ref 2: Steven Bird, Ewan Klein & Edward Loper (2009) Natural Language Processing with Python, O'Reilly, 978-0596516499

Ref 3: Andreas C. Muller (2016) Introduction to Machine Learning with Python, O'Reilly, 9781449369415

Ref 4: Paul Azunre, Transfer Learning for Natural Language Processing, Manning, 978-1617297267

PREREQUISITE(S)/ RECOMMENDATIONS: 3250 Foundations of Data Science or equivalent course/work experience in programming in Python.

Recommended:

You should bring a laptop robust enough to run current operating systems and connect to WiFi in class (if using a work laptop). [Specifications: System Type: 64 bit operating system, X64-based processor; Windows 7/8/10, Mac OS X or Linux; Processor: Intel ® i5-3230M CPU @ 2.6 GHz or better; Installed Memory (RAM): 8 GB or more].

You will use a free Google account to access Google Drive and Google Colaboratory. A basic knowledge of these platforms is recommended.

CERTIFICATE(S): This course is applicable to the following SCS Certificates

Advanced Artificial Intelligence

COURSE DESCRIPTION:



Natural language processing (NLP) can help your enterprise improve strategy and operations through deeper understanding of its products, processes and customers. In this 12-week course, you'll learn to process natural language and textual data and derive actionable insights using state-of-the-art NLP algorithms and techniques. You'll apply advanced machine learning and deep learning techniques to explore real-world problems related to textual-data processing, and create language-aware solutions. You'll also use open source production-ready Python libraries to build your own chatbot application.

Learners have the option to attend this course in-class or remotely. Each in-class session will be livestreamed through high quality audio/video interactive webinars for learners attending online.

LEARNING OUTCOMES:

- Apply NLP techniques to better understand and improve the way you do business.
- Solve complex real-world problems related to textual data.
- Build language-aware solutions using advanced machine learning and Deep Learning techniques.
- Use Python to perform text classification and topic modeling, graph analysis, and visual steering.
- Create your own NLP application.

ASSOCIATIONS: This course is not recognized by an Association.

<https://learn.utoronto.ca/programs-and-courses/association-partnerships>

COURSE FORMAT:

Course Format	Definition
Online with Real Time Meetings	Instructor-led online learning that may include: -Synchronous learning activities such as webinars which are also recorded -Asynchronous learning activities -Peer to peer interaction, group work -Pre-defined schedule with firm start and end dates

QUERCUS and ONLINE RESOURCES



Quercus is a learning management engine, and is the University of Toronto's main online teaching and learning environment. It is web-based software which gives you and your instructor a shared learning space online to receive and exchange course content as well as to communicate using a range of tools. More information about online learning in Quercus can be found here: <https://help.learn.utoronto.ca/hc/en-us/sections/115000462414-Online-learning-in-Quercus>

COURSE PLAN:

MODULES	DATE	TOPIC/LESSONS/WEBINARS	READINGS, ASSIGNMENTS, ACTIVITIES, ETC.
1	16-Sep-2024	Introduction to Natural Language Processing	• Ref 1, Chap 1 • Ref 2, Chap 1
2	23-Sep-2024	Text Corpora and Preprocessing	• Ref 1, Chap 2-3 • Ref 2, Chap 2-3
3	30-Sep-2024	Fundamentals of Machine Learning	• Ref 3, Chap 1-5
4	14-Oct-2024	Text Vectorization and Feature Engineering	• Ref 1, Chap 4
5	21-Oct-2024	Supervised and Unsupervised learning on text	• Ref 1, Chap 5-6 • Ref 2, Chap 6
6	28-Oct-2024	Context Aware Language Modeling	• Ref 1, Chap 7 • Ref 2, Chap 5 & 8
7	04-Nov-2024	Text Visualization & Graph Analysis	• Ref 1, Chap 8-9
8	11-Nov-2024	Scaling with Multiprocessing and Spark	• Ref 1, Chap 11
9	18-Nov-2024	Deep Learning on text data	• Ref 1, Chap 12 • Ref 4, Chap 4-6
10	25-Nov-2024	Transfer Learning for natural language processing	• Ref 4, Chap 1-3, 7-8
11	02-Dec-2024	LLM and Gen AI for Text	
12	09-Dec-2024	Team project Presentations	

GRADING AND EVALUATION:



**UNIVERSITY OF TORONTO
SCHOOL OF CONTINUING STUDIES**

Assignments	70%
Term Project	20%
Participation	10%

A note about Participation Marks:

Participation marks are for attendance and, more importantly, constructive contributions to class discussion on a regular basis.

Learners can expect to receive feedback and marks, if applicable, before the course end date, for all their submitted assignment(s) and test(s) other than the final exam, project or course paper. However, it is the sole responsibility of learners to make sure that they do get these marks from their Instructor and have all related questions answered before the course ends.

MISSED TEST/ASSIGNMENT GUIDELINES

If you miss a test or assignment please work directly with your instructor to make alternate arrangements. There may be penalties for missed or late assignments and tests.

SCS GRADING SCALE:

A 80% to 100% Excellent

B 70% to 79% Good

C 60% to 69% Adequate

D 50% to 59% Marginal

FX Less than 50%

INC Incomplete

DNW Did not write

FINAL GRADE:

To view your final grade, please login to “My Access – Student Login” at:
<https://learn.utoronto.ca/login>. Please note that your final grade will not be posted on Quercus.

More information regarding Academic Policies and Guidelines is located here:
<https://help.learn.utoronto.ca/hc/en-us/sections/207314307-Academic-Policies-and-Guidelines>

CODE OF STUDENT CONDUCT AND CODE OF BEHAVIOUR ON ACADEMIC MATTERS:



UNIVERSITY OF TORONTO
SCHOOL OF CONTINUING STUDIES

All School of Continuing Studies learners are required to comply with the University of Toronto Academic Policies including, but not limited to the Code of Student Conduct and the Code of Behaviour on Academic Matters.

Information regarding University of Toronto Academic Policies can be reviewed here:
<https://help.learn.utoronto.ca/hc/en-us/articles/235279047-Academic-Policies-and-Student-Code-of-Conduct>

ACADEMIC HONESTY:

Course work that is not appropriately cited may be in violation of the Code of Behaviour on Academic Matters (see above).

For guidelines about plagiarism and properly citing your sources, please visit:

<https://help.learn.utoronto.ca/hc/en-us/articles/115006427548-Academic-Honesty>

AUDIO/VIDEO RECORDINGS:

You are not permitted to record lectures without the written consent of your instructor(s).

ACADEMIC ACCOMMODATIONS:

If you require accommodation for a disability, please contact Enrolment and Learner Services at 416-978-2400, email scs.accessibility@utoronto.ca or fill out the form at the following link to arrange this service. <https://learn.utoronto.ca/help/forms-and-applications/accommodation-request-form>